(a) Ordered stages and cumulative cutoffs

Base — 45 ms

Diversified-mask feasible construction

Global — 60 ms

marginal repricing

CG — 70 ms

Same CG

CG-focused masks and vaild sharing

Remask — 78 ms

Remaining-demand masks and one-resource repair

Pair — 84 ms

Two-resource ruin-and-recreate

τ= 87 ms | Search cutoff: 84 ms; reserved tail: 3 ms | Elapsed time includes input parsing. | Cutoffs are cumulative, not stage durations. | Full = Base + Global + CG + Remask + Pair

(b) Shared commit-or-discard gate

Launch only if prerequisites hold and $\tau_{in} < t_j$.

Next attempt / stage

Skip expired stages

Initialize $I \leftarrow I_0$

Empty, complete, feasible

Stage / candidate controller

Eligible work remaining?

Yes

No

Build isolated candidate C

I stays unchanged during search

Common transactional gate

Complete: $\text{complete}(C)$

Feasible: $V(C) = I$

Timely: $\tau_{done}(C) < t_j$

Improving: $F(C) > F(I)$

τ_{done} is measured after validation.

PASS ALL

Commit: $I \leftarrow C$

FAIL ANY

Discard C; keep I

Final validation & serialization

Reserved 3-ms tail

$\text{complete}(I) \wedge V(I) = I$?

Yes

Serialize and output I

No

Restore $I \leftarrow I_0$

Serialize and output I_0

A complete feasible incumbent is retained throughout refinement.

100-ms deadline: evaluation only.